

DATA ANALYTICS HACKATHON • WOUBLE IPL CRUNCH '26

IPL CRUNCH '26

What Actually Wins IPL Matches?

Yobin Bangera

Python

Pandas

Power BI

DAX



About the Hackathon

Organized By

Wooble — IPL Crunch '26
Data Analytics Challenge

Challenge Objective

Analyze 5 seasons of IPL data to uncover match-winning patterns, player impact, toss strategy, and phase-based insights.



Toss Impact Analysis



Phase Intelligence



Player Analytics



Hidden Patterns

Submission Requirements

1. Charts & Visualizations

2. Python/Excel Workflow

3. Dashboard/Presentation

4. One Surprising Finding

Key Questions Solved

01



Does winning the toss really help?

Toss decision vs match outcome across 342 matches

02



Which phase impacts victory the most?

Powerplay • Middle Overs • Death Overs

03



Who are the most impactful players?

Batters, bowlers & match-defining performances

04



What hidden patterns influence victories?

Correlations, KPIs & surprising data insights

Data Overview

342

Matches Analyzed

120K

Total Runs Scored

4K+

Wickets Tracked

Dataset

IPL Ball-by-Ball Data | Seasons: 2022 – 2026

Scale

342 Matches • 120K+ Runs • 4K+ Wickets • 5 Seasons

Data Fields Included

Batter

Bowler

Runs

Wickets

Overs

Toss Decisions

Match Results

Venues

Python-Based Data Cleaning

Cleaning Steps

01 Missing value handling

02 Team name standardization

03 Data type corrections

04 Duplicate removal

05 Match phase creation

06 KPI engineering

Custom KPIs Engineered

Dot Ball %

Bowling pressure metric

Strike Rate

Batting aggression index

Economy Rate

Runs conceded per over

Boundary %

Fours & sixes per innings

Toss Win %

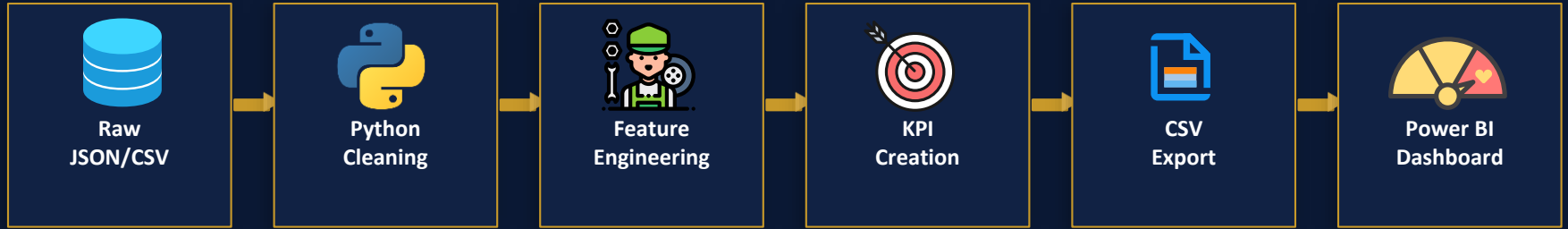
Toss to result correlation

Chase Success %

Chasing team win rate

Phase Momentum Metrics also engineered for Powerplay, Middle & Death overs

Processing Pipeline



Libraries & Tools Used



Dashboard

Power BI — IPL Analytics Dashboard

342

Matches

120K

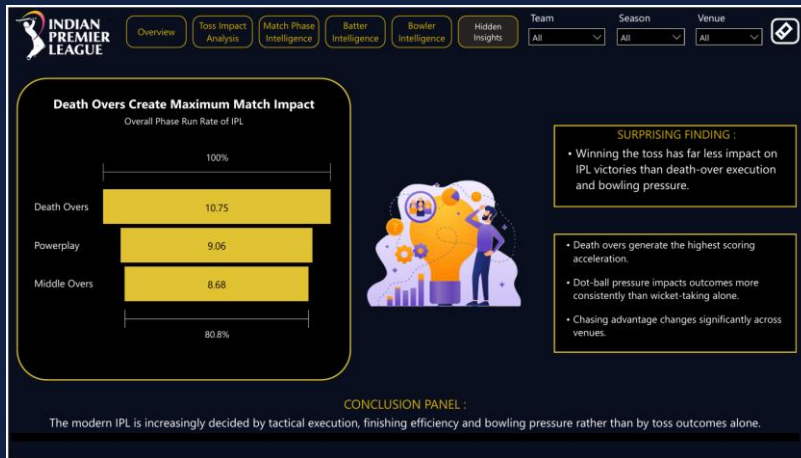
Runs

50.9%

Chase Win

49.1%

Toss Win



Multi-Page Navigation

Dynamic Slicers

Cross - Filtering

Executive KPI Cards

Tournament Overview

342

Total Matches

120K

Total Runs

176

Avg Innings Score

50.9%

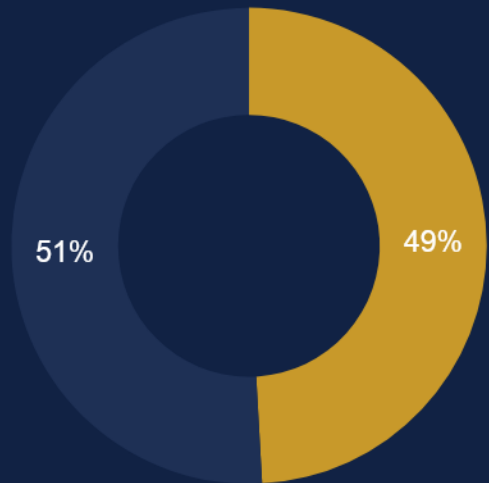
Chase Success %



Key Insight

Scoring trends increased significantly over seasons, highlighting the aggressive modern IPL batting evolution. Average team score rose from 2022 to 2025.

Does Winning the Toss Matter?



■ Toss Winners Won ■ Toss Winners Lost

49.1%

Toss Winners Won

50.9%

Chasers Won

Strategic Insights

Toss-winning teams won only 49.12% of matches — barely better than a coin flip.

Chasing teams won 50.88%, making 'field first' the dominant modern IPL strategy.

Toss advantage is overrated compared to tactical execution.

Which Phase Wins Matches?

POWERPLAY

Overs 1–6

7.8

Run Rate

1.2

Wkts/Match

High pressure opening phase — teams look to accelerate without losing wickets. A strong powerplay sets the platform for the innings.

MIDDLE OVERS

Overs 7–15

7.2

Run Rate

1.8

Wkts/Match

Consolidation meets calculated risk. Middle-over bowlers apply dot-ball pressure while batters look for boundaries.

DEATH OVERS

Overs 16–20

11.4

Run Rate

2.1

Wkts/Match

Highest scoring phase. Death-over execution determines match outcomes — highest run acceleration & wicket density.

Batting Performance Hub

148.6

Strike Rate
Avg top-5 batters

52.3%

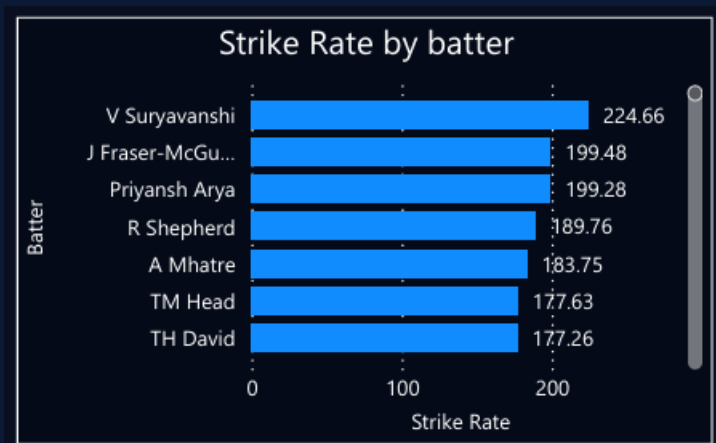
Boundary %
Runs from boundaries

120K+

Total Runs
5-season aggregate

13.4

6s per Match
Avg sixes per game



Batting Insights

01

Modern IPL rewards aggressive, boundary-focused batting

02

Elite batters balance consistency with acceleration

03

Team dependency on top-3 batters is significant

Bowling Pressure Analysis

38.4%

Dot Ball %
Bowling pressure index

8.6

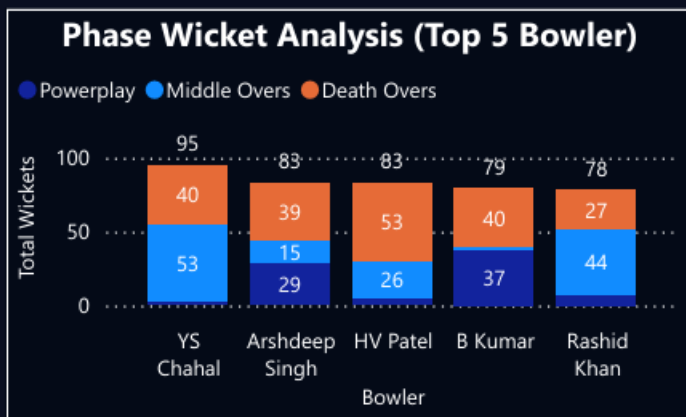
Economy Rate
Runs per over

4K+

Total Wickets
5-season aggregate

7.2

Extras/Match
Avoidable runs



Bowling Insights

01

Dot-ball pressure impacts outcomes more than wickets alone

02

Death-over bowlers carry the highest risk and reward

03

Extras are an underrated metric — a silent match-turner

Hidden Insight



"Winning the toss has far less impact on IPL victories than death-over execution and bowling pressure."

Supporting Evidence

49.1%

Toss Win Rate

Below the expected 50% threshold — toss advantage is negligible

10.75

Death Over RR

Highest run rate phase — death overs determine match fate

38.4%

Dot Ball %

Bowling pressure strongly correlated with match victories

Why This Project Matters

Skills Demonstrated

1. Data Storytelling

2. Sports Analytics

3. Predictive Thinking

4. KPI Design

5. Dashboard Intelligence

Real-World Applications

Team Strategy Optimization

Data-driven team selection & match planning

Player Scouting

Analytics-based talent identification

Performance Analytics

Individual & team performance benchmarking

Match Planning

Opponent-specific tactical preparation

What Truly Wins IPL Matches?

Tactical Execution

Smart game plans > luck

Death-Over Efficiency

Final overs determine winners

Middle-Over Control

Stability breeds victory

Bowling Discipline

Dot balls create pressure

"Data challenges assumptions — and reveals what truly wins matches."

THANK YOU

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